INCH-POUND
A-A-59192
20 May 1998
SUPERSEDING
MIL-T-17370C
10 March 1989

COMMERCIAL ITEM DESCRIPTION

TUBE AND FLUE CLEANER, BOILER, INTERNAL, AIR MOTOR DRIVEN

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

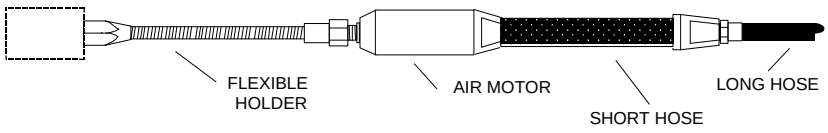
1. SCOPE

1.1 Tube and flue cleaners are outfits containing a compact tubular air motor, air hose, universal joints and flexible brush holders for removing deposits from the internal surfaces of boiler tubes and flues. The outfits are assembled with expanding brushes or other cleaning heads to traverse the interior bore of boiler tubes. These tubes can contain bends as small as 2-1/4 inch radius.

2. CLASSIFICATION

2.1 The tube and flue cleaners shall be of the following types:

Type I - Cleaning fire tube boilers
Type II - Cleaning water tube boilers



TYPICAL CLEANER ASSEMBLY
BRUSH/CLEANER HEAD NOT SHOWN

Beneficial comments recommendations, additions, deletions, clarifications etc. and any other data which may improve this document should be sent to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160.

AMSC N/A FSC 4940

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3. SALIENT CHARACTERISTICS

3.1 Component parts. Each tube and flue cleaner for fire- or water-tube boilers shall be constructed so that its' component parts can be joined together in proper order of assembly, using standard machine thread and coupling pipe threads in accordance with the tables herein, into one complete unit, which shall consist of the parts shown below. Kits for type II cleaners size 2 and below are intended for use with expanding brushes bought to A-A-59193. Neither these brush assemblies or any other cleaning head should be included in these type II cleaning outfits unless specified in the contract.

(a) Type I cleaner:

Air Motor (suitable for powering swing frame cutter, knocker, or vibrator head assemblies).
Operating Hose.
Universal Joint.
Wrench for Rotor.
Repair parts list and operating instructions toolbox.
4-6 blade expanding scraper assembly.

(B) Type II cleaner:

Air motor with short hose and coupling attached.
Universal joint (when required).
Flexible holder.
Operating hose complete with couplings attached.
Wrench for rotor.
Wrench for flexible holder retainer.
Repair parts list and operating instructions.
Toolbox (except size 4).

3.2 Operating pressure. The tube and flue cleaners shall effectively clean the walls of boiler tubes when operated with compressed air supplied at gauge pressure between 80 and 100 pounds per square inch (lb/in^2).

3.3 General construction. The tube and flue cleaners specified herein shall be of two types (I and II) and shall perform the functions of removing rust, scale, and other normal boiler deposits from the walls of tubes in fire- and water-tube boilers. Parts subject to wear, breakage, or distortion shall be accessible for easy adjustment and repair with no special tools required.

3.4 Standard sizes. The type I tube and flue cleaner shall be sized to clean the walls of straight fire tube boiler tubes as specified in table I. The type II tube and flue cleaner shall be sized to accommodate the brushes used to clean the interior walls of boiler water tubes as specified in table II.

3.5 Threads. Dimensions of all threads, tolerances and gauges for determining interchangeability shall comply with FED-STD-H28. Tapered pipe threads shall be used where applicable. Thread class selection shall be unified coarse (UNC) class 2A-2B for general use and UNC class 3A-3B for tighter fits.

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3.6 Performance.

3.6.1 Hose performance. The hose with couplings attached shall be tested to a hydrostatic pressure of 250 lb/in^2 and maintained for 2 minutes without exposing imperfections or leaks in either the hose or the couplings.

3.6.2 Motor performance. Motors shall have performance characteristics superior or equal to those shown in table III for type I cleaners or table IV for type II cleaners.

3.7 Hose. Hose supplied shall be rubber with yarn or fabric reinforcement designed for pneumatic tool service meeting ZZ-H-500.

3.7.1 Short hose. The short hose for size 1-1/2 units and larger shall be wire reinforced externally in accordance with the manufacturer's standard practice. Where more than one type of hose assembly is offered for a particular motor, the armored or heavier ply type shall be provided unless a lesser design hose is specified in the contract. The short hose (type II) for connecting to the motor shall be fitted with a female pipe thread coupling on the air supply end for attachment to the long operating hose.

3.7.2 Long hose. The long hose for connecting the cleaning unit to the compressed air line shall be fitted at each end with a male threaded coupling. Unless otherwise specified hose for type I cleaners shall be 50 feet long, hose for type II cleaners shall be 25 feet long.

3.7.3 Couplings. Hose coupling sizes shall be as specified in tables V and VI.

3.7.4 Flexible holders. Flexible holders shall be as specified in table VII.

4. REGULATORY REQUIREMENTS

4.1 The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market acceptability.

5.2.1 A market tested item is required to assure reliability and cleaning effectiveness without costly performance testing. The company producing the item must have at least 5 years experience designing and producing pneumatic and electric boiler tube cleaning equipment. Service proven designs and established Q.A. programs are essential to minimize cleaner components lodging in boiler tubes or otherwise damaging tubes that will necessitate costly boiler repairs. Also, failure of portable pneumatic tools, such as these, in the confined space of boiler drums, creates a safety hazard to personnel. Only companies who specialize in these tools and can so demonstrate by producing catalogs, statistics of items sold or other documentation should respond.

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5.2.2 Test and inspection. Bid samples may be required when necessary to assure product quality and interchangeability with brush assemblies (A-A-59193) and other cleaner components.

6. PACKAGING

6.1 Packaging shall meet the standard practice requirements of ASTM D 3951 unless special preservation, packaging, packing, labeling, and marking are specified in the contract or order.

7. NOTES

7.1 CID-based parts identification number. The following part identification number procedure is for government purposes and does not constitute a requirement for the contractor. This example describes a part numbering system for A-A-59192.

Example: A-A-XXX-2253AN5
2"OD watertube cleaner
7/16x14 rotor thd; 3/8 NPT
hose cplg; 2.5" flex. hldr;
no cleaning head and 25' hose.

Type: 1 = type I firetube
2 = type II watertube

Size: 1 = 1" OD tube
A = 1.125" OD tube
B = 1.25 " "
C = 1.375 " "
D = 1.5 " "
E = 1.75 " "
2 = 2" OD tube
F = 2.25 OD tube
G = 2.5 ID tube
3 = 3" ID tube
4 = 4" ID tube
H = 4.5 ID tube
I = 4.75 ID tube

Rotor Thd (Male): 1 = 7/32 x 24
2 = 1/4 x 20
3 = 5/16 x 18
4 = 3/8 x 16
5 = 7/16 x 14
6 = 9/16 x 12
7 = 5/8 x 11
8 = 3/4 x 10
9 = 7/8 x 9
0 = 1-1/8 x 12

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Coupling pipe thread size:	1 = 1/4 NPT 3 = 3/8 NPT 4 = 1/2 NPT 6 = 3/4 NPT 8 = 1 NPT
Flexible Holder Length: (spring length)	N = none provided A = 2.5" B = 2.75" C = 4.125" D = 5" E = > 5.25"
Cleaning head type:	N = None provided C = Single pin cone cutter A = Four arm cutter assy S = Expanding scraper B = Expanding brush F = Swing frame brush K = Knocker/vibrator X = All other Multi pin cutter assemblies
Long hose Length:	3 = 15 ft 5 = 25 ft 7 = 35 ft 0 = 50 ft

7.2 Source documents.

7.2.1 ASTM standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2859.

7.2.2 Federal Standard FED-STD-H28 and Federal Specification ZZ-H-500 may be obtained by sending a request for copies to the Defense Automated Printing Service, Attn: DODSSP, 700 Robbins Avenue, BLDG. 4D, Philadelphia, PA 19111-5094.

7.3 National stock numbers (NSNs). The following is a list of NSNs associated with air motor driven tube cleaning outfits. The list may not be indicative of all possible NSNs that could be associated with the CID.

NSN	Type	Size
4940-00-224-7777	II	1
4940-00-224-7778	II	1-1/4
4940-00-224-7779	II	1-1/2
4940-00-254-4780	II	1-1/2SH
4940-00-224-7780	II	2

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4940-00-516-5614	II	2*
4940-00-977-3431	II	2-3/4*
4940-00-964-1704	II	2.5
4940-00-224-7781	II	3
4940-00-964-1705	II	3*
4940-00-224-7782	II	3-1/4
4940-00-079-4927	II	3-1/4*
4940-00-242-8639	II	4
4940-00-410-2619	II	4-3/4 ID

* - With 4 arm expanding cutter

4940-00-964-0821	I	1-1/2
4940-00-516-5614	I	1-1/2*
4940-00-237-9879	I	1-3/4
4940-00-516-5615	I	2
4940-00-516-6191	I	2.5
4940-00-516-5617	I	3
4940-00-964-0822	I	4

* - With solid cutter vice expanding scraper

7.4 Source of supply. These products are currently available commercially as shown:

INTOOL Incorporated
Industrial Energy Products Div.
P.O. Box 268
Springfield, OH 45501

Elliott Co.
Industrial Tool Division
1760 Tuttle Ave.
Dayton, OH 45401

Thomas C. Wilson Co.
21-11 44th Ave.
Long Island City, NY 11101-5088

TABLE I. Size of firetube units.

Cleaning unit size designation	Outside diameter of boiler tube (inches)	Minimum inside diameter of boiler tube (inches)
1-1/2	1-1/2	1-1/4
2	2	1-3/4
2-1/2	2-1/2	2-1/4
3	3	2-5/8
3-1/2	3-1/2	3-1/8
4	4	3-5/8

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TABLE II. Size of watertube units (type II).

			Dimensions of curved sections		
Cleaning unit size designation	Outside diameter of boiler tube (inches)	Wall thickness of boiler tube (inches)	Radius (inches)	Arc (degrees)	Maximum crush of tube (inch)
1	1	0.095	5	75	0.050
1 SH 1/	1	0.109	2-1/4	180	0.0625
1-1/8	1-1/8	0.109	5	75	0.050
1-1/4 spec.	1-1/4	0.109	5	75	0.062
1-1/4	1-1/4	.095 to .134	4	90	0.0625
1-1/4 SH 1/	1-1/4	0.120	2-1/2	180	0.0625
1-1/4 SH	1-1/4	0.120	2-1/2	180	0.0625
1-3/8	1-3/8	0.120	6	75	0.062
1-1/2	1-1/2	0.120	6	75	0.075
1-1/2 SH 1/	1-1/2	.105 to .120	2-1/2 to 5	75 to 180	0.0625
1-3/4	1-3/4	0.134	6	75	0.100
2	2	0.134	6	75	0.120
2 SH 1/	2	0.148	5	90	0.120
2-1/2	2-1/2	0.165	12	90	0.120
3	3	0.165	12	75	0.125
3A 2/	3	0.250	12	75	0.125
	4	0.250	15	75	0.125
3-1/4 2/	3-1/4	0.250	18	75	0.125
	4-1/2	0.343	16	75	0.125
3-1/4A	3-1/4	0.375 max	10	62	0.125
4	4	0.375 max	15	120	0.125
4	4	0.375 max	10	53	0.125
4-1/2	4-1/2	0.343 max	15	48	0.125
4-1/2	4-1/2	0.343 max	12	90	0.125

See footnotes at top of next page.

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1/ Special units shall be constructed for special superheater (SH), side or rear wall, roof or economizer tube configurations, and header conditions.
2/ Constructed for both tube sizes specified.

TABLE III. Performance characteristics of motors for firetube units.

Cleaning unit size designation (inches)	Performance at specified torque		
	Specified torque (foot-pounds) (minimum)	Speed (r/min) (minimum)	Air consumption (ft^3/min per) horsepower (maximum)
1-1/2	0.09	4000	200
2	0.40	4000	175
2-1/2	1.50	3000	125
3	2.75	3000	100
3-1/2	3.50	3000	75
4	3.50	3000	75

TABLE IV. Performance characteristics of motors for watertube units.

Cleaning unit size designation (inches)	Performance at specified torque		
	Specified torque (foot-pounds) (minimum)	Speed (r/min) (minimum)	Air consumption (ft^3/min per) horsepower (maximum)
1	0.03	4000	300
1 SH	0.03	4000	300
1-1/8	0.04	4000	275
1-1/4	0.06	4000	250
1-1/4 SH	0.06	4000	250
1-3/8	0.07	4000	225
1-1/2	0.09	4000	200
1-1/2 SH	0.09	4000	200
1-3/4	0.12	4000	175
2	0.14	4000	150
2 SH	0.14	4000	150
2-1/2	0.19	4000	125
3	1.20	4000	100
3A	0.65	4000	100
3-1/4	0.65	4000	100
3-1/4	1.20	4000	100
4	2.00	4000	60
4-1/2	2.50	4000	50
5	3.00	4000	40
6	4.00	3500	40
8	6.00	3500	40

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TABLE V. Hose and coupling sizes - type I cleaning units.

Cleaning unit size designation (inches)	Nominal inside diameter of hose (inch)	Coupling pipe thread size male (inch)
1-1/2	3/8	3/8
2	1/2	1/2
2-1/2	3/4	3/4
3	1	1
3-1/2	1	1
4	1	1

TABLE VI. Hose and coupling sizes - type II cleaning units.

Cleaning unit size designation (inches)	Nominal inside diameter of hose (inches)	Length over couplings (short hose only) (inches)	Coupling pipe thread size (inch)
1 and 1 SH	1/4	5 to 5-1/2	1/4
1-1/8	1/5	5-1/4 to 5-3/4	1/4
1-1/4 and 1-1/4 SH	1/4	5-3/8 to 5-7/8	1/4
1-3/8	3/8	5-3/8 to 5-7/8	3/8
1-1/2 and 1-1/2 SH	3/8	5-1/2 to 6	3/8
1-3/4	3/8	5-5/8 to 6-1/8	3/8
2 and 2SH	3/8	5-3/4 to 6-1/4	3/8
2	1/2	5-3/4 to 6-1/4	1/2
2-1/2	1/2	13-1/4 to 14-1/4	1/2
3	1	9-1/4 to 10	1
3	3/4	13-1/4 to 14-1/4	3/4
3A	1/2	13-1/4 to 14-1/4	1/2
3-1/4	1/2	13-1/4 to 14-1/4	1/2
4	3/4	13-1/4 to 14-1/4	3/4
4-1/2	1	13-1/4 to 14-1/4	1

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TABLE VII. Flexible holder for type II cleaner.

Cleaning unit size designation	Threads cleaning device end			Length	Overall length (minimum) (inches)
	Size (outside diameter) (inch)	Threads per inch 1/	Type		
1	5/16	18	Female	5/16 to 3/8	3-1/2
1 SH	5/16	18	Female	5/16 to 3/8	8
1-1/8	5/16	18	Female	5/16 to 3/8	3-5/8
1-1/4	5/16	18	Female	5/16 to 3/8	3-5/8
1-1/4 SH	5/16	18	Female	5/16 to 3/8	8
1-3/8	5/16	18	Female	5/16 to 3/8	3-5/8
1-1/2	5/16	18	Female	5/16 to 3/8	4-1/4
1-1/2 SH	5/16	18	Female	5/16 to 3/8	8
1-3/4	7/16	14	Female	3/8 to 7/16	6
2	7/16	14	Female	3/8 to 7/16	6
2 SH	7/16	14	Female	3/8 to 7/16	9
3	9/16	12	Female	7/16 to 1/2	9-1/8
3A	9/16	12	Male	7/16 to 1/2	9-1/8
3-1/4	9/16	12	Male	1/2 to 9/16	9-1/8
4	9/16	12	Male	1/2 to 9/16	10-9/16
4-1/2	7/8	9	Male	9/16 to 5/8	10-9/16

1/ Tolerance UNC 2A-2B applies; UNC 3A-3B acceptable.

MILITARY INTERESTS:

Custodians:
Army - GL
Navy - SH
Air Force - 99

Preparing activity:
Navy - SH
(Project 4940-0719)

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